

Revised Load Interconnection System Impact Study Report

For Queue Project Q19-1294

Merriam 138 kV
September 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1294. The Interconnection Customer proposes a 1200.0 MW large load (data center) facility interconnecting at the Merriam 138 kV substation in the Maryland area. The request was received on September 21, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$119,151,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Belt to Ss3145 138 kV	Quincey to Merriam 138 kV	825.3	152	100.0
Eckert Sta to Ss3145 138 kV	Merriam to Quincey 138 kV	652.2	198	100.0
Merriam Substation to North Belt 138 kV	Merriam to Quincey 138 kV	637.0	194	100.0
Merriam to Quincey 138 kV	Marriott Hill to Fulton Substation 138 kV	629.5	218	100.0
Billingsley to Eckert Sta 138 kV Line	Quincey to Merriam 138 kV	603.2	221	100.0
Marriott Hill to Billingsley 138 kV	Merriam to Quincey 138 kV Line	573.1	237	100.0
Fulton to Marriott Hill Substation 138 kV	Merriam to Quincey 138 kV	562.1	244	100.0
Air Products 1 to Fulton 138 kV	Quincey to Merriam 138 kV	297.9	198	46.9
Air Products 1 to Highland Avenue 138 kV L	Quincey to Merriam 138 kV	287.8	214	46.9
Highland Avenue to Shingletown 138 kV line	Quincey to Merriam 138 kV L	287.2	218	46.9
Mitchellville Substation to Buena Vista 138 kV	Priest Bidge Substation to Levitt 138 kV	226.1	198	33.3
Levitt to Buena Vista Substation 138 kV L	Buena Vista to Mitchellville 138 kV L	174.8	209	24.3
Levitt to Webster Street 138 kV	Buena Vista to Levitt 138 kV L	157.5	186	22.6
Mitchellville to Quincey 138 kV	Levitt to Priest Bridge 138 kV line	149.4	289	33.3
Camp Gary to Southwest 138 kV	Steed Road to Tulu 138 kV	144.6	175	7.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fort Slocum to Carrollton-Josey Lane 138 kV line	Seek Lane to Flanagan 138 kV	137.2	219	5.7
Oak Grove to Ritchie 138 kV	Mt Lassen to Central Avenue 138 kV L	136.6	249	17.1
Priest Bridge to Levitt 138 kV	Quincey to Central Avenue 138 kV Line	132.1	397	41.8
Tokoma to Fort Slocum 138 kV L	Flanagan to Seek Lane 138 kV	130.2	189	5.7
Camp Gary to Dryden 138 kV	Steed Rd to Tulu 138 kV	129.2	183	7.6
Cal Sierra Development to Oak Grove 138 kV line	Mt Lassen to Central Avenue 138 kV L	128.6	243	13.6
Central Avenue to Quincey 138 kV	Priest Bridge to Levitt 138 kV Line	127.6	377	40.2
Mt Lassen to Seat Pleasant 138 kV	Ritcie to Oak Grove 138 kV	124.7	335	25.8
Mt Assen to Central Avenue 138 kV L	Ritchie to Oak Grove 138 kV	124.1	339	25.8
Hartford to Kingswood 138 kV	Ritchie to Oak Grove Substation 138 kV	121.2	320	15.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Belt to Ss3145 138 kV	Rebuild line with higher-capacity conductor	\$2,056,000	100% (\$2,056,000)
Eckert Sta to Ss3145 138 kV	Rebuild line with higher-capacity conductor	\$1,049,000	100% (\$1,049,000)
Merriam Substation to North Belt 138 kV	Rebuild line with higher-capacity conductor	\$9,208,000	100% (\$9,208,000)
Merriam to Quincey 138 kV	Rebuild line with higher-capacity conductor	\$1,917,000	100% (\$1,917,000)
Billingsley to Eckert Sta 138 kV L	Rebuild line with higher-capacity conductor	\$7,439,000	100% (\$7,439,000)
Marriott Hill to Billingsley 138 kV	Rebuild line with higher-capacity conductor	\$4,146,000	100% (\$4,146,000)
Marriott Hil Substation to Fulton 138 kV Line	Rebuild line with higher-capacity conductor	\$34,847,000	100% (\$34,847,000)
Air Products 1 to Fulton Substation 138 kV	Rebuild line with higher-capacity conductor	\$7,207,000	100% (\$7,207,000)
Air Products 1 to Highland Avenue 138 kV L	Rebuild line with higher-capacity conductor	\$4,644,000	100% (\$4,644,000)
Highland Avenue to Shingletown 138 kV line	Rebuild line with higher-capacity conductor	\$1,064,000	100% (\$1,064,000)
Buena Vista to Mitchellville 138 kV	Rebuild line with higher-capacity conductor	\$9,891,000	100% (\$9,891,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Levitt to Buena Vista Substation 138 kV L	Rebuild line with higher-capacity conductor	\$4,486,000	100% (\$4,486,000)
Levitt to Webster St 138 kV	Rebuild line with higher-capacity conductor	\$2,330,000	100% (\$2,330,000)
Quincey to Mitchellville Substation 138 kV line	Rebuild line with higher-capacity conductor	\$2,918,000	100% (\$2,918,000)
Southwest Substation to Camp Gary 138 kV	Rebuild line with higher-capacity conductor	\$1,336,000	100% (\$1,336,000)
Carrollton-Josey Lane to Ft Slocum 138 kV line	Rebuild line with higher-capacity conductor	\$964,000	98% (\$941,000)
Oak Grove to Ritchie 138 kV	Rebuild line with higher-capacity conductor	\$1,500,000	100% (\$1,500,000)
Levitt to Priest Bridge 138 kV	Rebuild line with higher-capacity conductor	\$2,255,000	100% (\$2,255,000)
Tokoma to Fort Slocum 138 kV L	Rebuild line with higher-capacity conductor	\$2,237,000	98% (\$2,185,000)
Camp Gary to Dryden 138 kV	Rebuild line with higher-capacity conductor	\$1,921,000	100% (\$1,921,000)
Cal Sierra Development to Oak Grove 138 kV	Rebuild line with higher-capacity conductor	\$2,611,000	100% (\$2,611,000)
Central Avenue to Quincey 138 kV	Rebuild line with higher-capacity conductor	\$9,895,000	100% (\$9,895,000)
Mt Lassen to Seat Pleasant 138 kV	Rebuild line with higher-capacity conductor	\$338,000	100% (\$338,000)
Mt Assen to Central Avenue 138 kV L	Rebuild line with higher-capacity conductor	\$789,000	100% (\$789,000)
Hartford to Kingswood 138 kV	Rebuild line with higher-capacity conductor	\$2,200,000	99% (\$2,178,000)
	Total allocated to project		\$119,151,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.